

it's not all electronics. Chips are made of metals, and metals come under metallurgy, with composition and organic structure. Thus it involves the sciences of chemistry, metallurgy, physics and electrical engineering too. Whatever is there, it's being controlled with the help of electronics. So it's no more just illumination. Lighting today is a multidisciplinary subject," explains Dr Prakash Barjatia, former director, MIT School of Energy & Lighting, Pune, and chairman, Mumbai State Centre, Indian Society of Lighting Engineers.

LED *per se* is not a light source. It's a semiconductor diode that emits light and the light is controlled by a controller. Without the controller, it can't work. Then comes the science of physics in heat dissipation. Whatever heat is generated, if it's not dissipated properly, the module will not perform effectively.

"Very few companies make LEDs in India. On the other hand, LED drivers are designed and manufactured by many companies in India," informs Dr Barjatia.

Skills required

LED lighting requires knowledge of all the sciences mentioned above. If you look at the basic concept of an LED, there is the role of electronics engineers. But if you consider an overall luminary where LED or some other light source can be used, role of lighting designers and electrical engineers too comes into play. Proper lighting is more a role of creativity as it can do wonders by elevating your mood, for instance.

It also helps to have some knowledge of management. To sell a product in the market, one needs to know how LED is better than other lighting technologies. LED is energy-efficient but there are other such sources as well.

"In India, the industry is limited to few people who keep on switching from one company to another. There is not much of value addition done year by year to this industry. The only way to do that is by infusing new blood into this industry. Without new ideas and

LED light fundamentals

Typically, LED lights package arrays of many individual LED devices onto printed-circuit boards along with power and driver/control circuits and suitable heat-sinks. For typical lighting applications, manufacturers use blue LEDs housed in a lens coated with yellow phosphor to produce white light. Lamp manufacturers combine these LED clusters with control circuitry into a variety of shapes.

LEDs are semiconductors that convert electricity into light. The driver converts line power into the appropriate voltage (typically, 2-4V DC for high-brightness LEDs) and current (generally, 200-1000 mA), and may also include dimming and/or colour-correction controls. Basically, it is a self-contained power supply that has outputs matched to the electrical characteristics of the LED or array of LEDs.

There are currently no industry standards, so understanding the electrical characteristics of the LED or array is critical in selecting or designing a driver circuit. Drivers may have more than one channel for separate control of different LEDs or arrays.

understanding of the vastness of this industry, things will never improve," says Mr Sood.

"Lighting can influence the whole working culture—offices, homes, hospitals, airports and industrial environments all need different kinds of lighting. The industry needs people from engineering background, architectural background (interior designers) and even physiologists because lighting has so much of influence on the human mind itself," he adds.

Where to study?

Jadavpur University offers a purely technology-oriented course. It offers M.Tech in illumination technology, which is a two-year full-time course. Also, there are some universities which have adopted illumination engineering as one of the elective subjects at the BE level. These include Manipal, Mumbai and Pune universities apart from Jadavpur University.

"As the industry needs knowledge of marketing too, on the lines of sector-specific MBA courses, we conceptual-

ised an MBA course in lighting technology and energy management. This course was launched in 2007 at MIT Group, Pune as a two-year full-time postgraduate programme. The success of this course can be gauged from the fact that till the last fourth batch, there have been more than 100 per cent placements; some candidates have even received more than three-four placement offers including overseas offers," informs Dr Barjatia.

The course structure of this programme is lighting technology (40 per cent), marketing management (40 per cent) and energy management (20 per cent). Lighting/energy management subjects taught are introduction to lighting technology, lighting and its measurement, light sources, luminaires, lighting applications, biological effects of light, lighting design, lighting management and energy management.

"With LED coming in a big way, everybody is feeling the need for LED lighting professionals. As a result, new courses are coming up. Some online courses are also available," adds Dr Barjatia.

At the global level, the industry is proactive and the community responds to the actions of the industry. The Lighting Research Center (LRC) in New York is one example. Part of Rensselaer Polytechnic Institute, it is a leading university-based research centre devoted to lighting. It also offers graduate education in lighting, including one- and two-year master's programmes and a Ph.D. programme. The Solid-State Lighting Program at LRC conducts necessary research and educational programmes to enhance LED lighting technology, overcome barriers to effective use, and help it to gain acceptance for general illumination purposes.

Each LED Lighting Institute of LRC is designed to teach industry professionals how to incorporate LEDs into lighting applications. It also includes updated technical content based on the latest industry developments. Through hands-on workshops